MOSFET

1. (a) $\frac{1}{C_{ox}} + \frac{1}{C_{dep}} = \frac{1}{C}$ | (b) $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$

2. $I_{DS} = \frac{1}{2}\mu \frac{W}{L} C_{ox}\left[(V_G - V_t)V_{DS} - \frac{V_{DS}^2}{2}\right]$.

3. I_{DS} (CLM) ① $= \frac{\beta}{2} C_{ox}(V_G - V_t)^2(1 + \lambda V_{DS})$

4. (Subthreshold current) → ② enhanced on-current.
$I_{sth} = I_s \exp\left(\frac{V_G - V_T}{n \cdot V_{thermal}}\right)$ → $\frac{KT}{q}$, $n \approx 1$

5. On resistance: $R_{on} = \frac{1}{\beta \cdot (V_G - V_T)}$

4.(b) $V_{thermal} = \frac{KT}{q} \ln 10$ (60 mV)

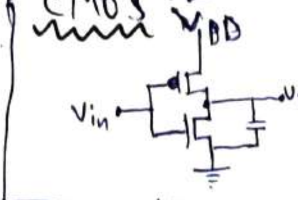
(6) Velocity saturation → ③
High field charge carriers lose V_{DS} dependence. $\left[V_d = \left(\frac{\mu}{1 + \frac{E}{E_c}}\right) \cdot E\right]$ $E < E_c$
$\left(E = \frac{V_{ds}}{L}\right)$ <early saturation>

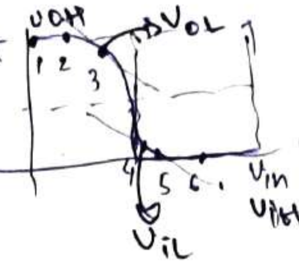
7. Hot injection & Gate induced Drain leakage. (High E field)
High K low doped drain.

8. DIBL Drain induced barrier leakage
<SOI, GAA, FinFET>.

9. Punch thru effect

CMOS :-





1. NMOS off, PMOS linear.
2. NMOS sat, PMOS linear
(3)(4) NMOS, PMOS sat (unstable)
5. NMOS linear, PMOS sat
6. NMOS linear, PMOS off.

(i) $I_{DSP} = -I_{DSN}$
(ii) $V_{GSP} = V_{in} - V_{DD}$
(iii) $V_{DSP} = V_{out} - V_{DD}$
(iv) $V_{GSN} = V_{in}$
(v) $V_{DSN} = V_{out}$.

(I) V_M = when NMOS sat + PMOS sat = ③.
(use I_{sat} equ to derive, two ① opposite side)

(II) Noise margin: (Negative gain)
output | input
NMH $= V_{OH} - V_{IH}$
NML $= V_{IL} - V_{OL}$

→ i) $V_{OH} = V_{DD}$, $V_{OL} = 0$, $V_{IL/H} = \left(\frac{\partial V_{out}}{\partial V_{in}} = -1\right)$ condition

(III) Skewing: $V_M = \frac{\sqrt{\beta_P}(V_{DD} - V_{tp}) + \sqrt{\beta_N}(V_{tn})}{\sqrt{\beta_P} + \sqrt{\beta_N}}$.

(IV) Power
i. Dynamic: $\alpha C_L V_{DD}^2 f$.
ii. Static: leakage $V_{DD} \cdot (I_{ST} + I_{GOL})$.
(iii) Short circuit: $V_{DD} \cdot I_{SC}$ (peak-to-peak).

Photolithography: ArF 193 nm laser, litho-etch-litho-etch (LELE) LFLE, SADP (self aligned double patterning).
→ vapour, adhesive (HMDS), spin coat, bake, (mask, irradiate), bake), (develop (TMAH), hard bake), etch.
→ $R = K_1 \frac{\lambda}{NA}$. (R needs to be decreased) (blue light preferred) λ fixed.
→ Etch (wet: Acid, HF) (Dry: Reactive ion, chemical, plasma). (just ion)
(NH_4OH, H_2O_2, deionized H_2O (heated))

Full fab process: ① Clean: organic, inorganic (just add HCl), oxide (HF treatment).
② Doping: ion implant (ion source (plasma), cyclotron acc, lens (magnetic quadrupole), e⁻ scanning).
diffusion (high temp process, $D_n \frac{dc}{dx}$, isotropic, low precision, deeper, less control)
③ Deposition: PVD: sputtering / e⁻ beam (bombard chemical source, nucleate on substrate).
(contamination) CVD: high temp. process, Reaction kinetics & fact needed, faster, vacuum need,
ALD: costlier, slower, precursors sequential, self limiting rxn, deposit-purge chain, slow.

④ Interconnect: ① Skin depth → eddy current due to AC. unwanted loss & dissipation.
MIT: thicker, wider conductor, reduce ρ.
② RC delay: (i) MIT: low K dielectric b/w wires. thicker wires help to reduce ρ. (RC=τ)
③ Electromigration: caused by current density alone, creates craters & hillocks.
MIT: Block diffusion / Barrier layer): Ta/TaN, confine mechanically (SiN, CoWP). use W vias.
④ Stress migration: due to heat expansion. use annealing to fix.

FEOL → transistor making BEOL → interconnect skeleton making

FINFET → good: improved electrostatic control, reduce leak current, overdrive current higher → BAD → design complexity, variability, thermal management, higher parasitic C.

SOI: improved subthreshold leakage, BOX (buried oxide) b/w substrate & ST.

FINFETS: Gate is wrapped around channel, increases surface area ↓ ρ (Resistance)
→ High K dielectric (HfO_2, ZrO_2) } challenges: self heating (try to confine it please)
→ interfacial layer (thin SiO_2) }

→ GAA